Computers and Electronics in Agriculture 212 (2023) 108021



Contents lists available at ScienceDirect

Computers and Electronics in Agriculture

journal homepage: www.elsevier.com/locate/compag





Navigating route planning for multiple vehicles in multifield agriculture with a fast hybrid algorithm

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ARTICLE INFO

Keywords:
Agriculture
Route planning
Multifields
Multimachine
Fast hybrid algorithm
Genetic algorithm
Tabu search
Ant colony optimisation

ABSTRACT

Optimisation of route planning is becoming increasingly valuable aspect in agriculture. This study focuses on Agricultural Route Planning (ARP) in multifield areas (with a specific entrance point), incorporating several heterogeneous agricultural machines. The aim of this research is to improve the route planning of (semi-)autonomous machines by producing more efficient route plans. The problem sets of ARP contain both medium and large numbers of tracks consisting of irregular and rectangular fields. This research proposes a Fast Hybrid Algorithm (FHA) to address this problem. FHA incorporates various combinatorial operators into its structure. The experimental results demonstrate that, compared to Tabu Search, (Improved) Genetic Algorithm and Ant Colony Optimisation, FHA can reduce the distance travelled by an average of 16.21%. Furthermore, the efficiency of FHA is also reflected in its running time, which saves up to 54.23% compared to the other methods.

1. Introduction

The logistics sector in agriculture is facing an increasing number of challenges. Optimisation of time and natural resources, such as fuel and water, is required to complete fieldwork in expanding farming fields due to concerns about climate change and demands for sustainable operations (Muhie, 2022). Environmental changes also affect the timeline for harvesting crops. Reducing expenses and carbon dioxide emissions further highlights the importance of optimising the path and usage of farming machines, including reducing the range travelled by harvesters (He et al., 2020).

Route planning is among the most critical tasks for managing agricultural machinery (Bochtis et al., 2014). Agricultural Route Planning (ARP) is an effective way to optimise farming operations, manage the logistics of agricultural machines, particularly their field movements to other farming sites. The ARP problem is focused on cutting down the length of trips taken by agricultural vehicles (hereafter called 'machines') for conducting fieldwork in all tracks (Utamima et al., 2019). Different adaptations have been implemented to achieve various objectives, including the minimisation of nonworking distance or time (Seyyedhasani et al., 2019) and specific fieldwork activities such as slurry application (Vahdanjoo et al., 2020), potato cultivation (Zhou et al., 2015) and row crop grain harvesting (Evans et al., 2020).

The majority of ARP studies use the nonworking distance concept, which Bochtis and Vougioukas (2008) first proposed. The optimisation of the nonworking distance is intended to minimise the distance travelled by the machine when not engaged in an agricultural activity. Adapting the constraint model proposed by Jensen et al. (2015) and Bochtis et al. (2015) used the model for orchard operations, followed by Conesa-Muñoz et al. (2016) to solve issues in killing weeds, and by Seyyedhasani and Dvorak (2018b) to minimise the duration of fieldwork. Valente et al. (2013) and Barrientos et al. (2011) applied the same concept to the planning of aerial coverage. Plessen (2019) reduced the nonworking distance in both full and partial field coverage in fields with barriers and an irregular shape. Edwards et al. (2017) analysed the situation of neutral machine operations, which involved no material flow into or out of the field (e.g., mowing, cultivation) while utilising a single machine and did not consider machine capacity.

ARP includes several constraints, such as the holding capacity for collected crops or products, the turning radius, the width of tracks, and fuel consumption. However, its capacitated application with different machines and multiple fields has yet to be extensively covered. While the focus has been on a single field in some studies (Khajepour et al., 2020; Conesa-Muñoz et al., 2016; Evans et al., 2020; Seyyedhasani and Dvorak, 2018b), there remains great potential for investigating ARP with several fields and multiple machines. Studies on sugarcane harvesting have addressed issues related to multiple crop fields (Kittil-ertpaisan and Pathumnakul, 2017; Worasan et al., 2020; Sethanan and Neungmatcha, 2016). For example, issues related to tractor scheduling,

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https://doi.org/10.1016/j.compag.2023.108021
Received 23 November 2022; Received in revised form 15 June 2023; Accepted 25 June 2023
Available online 27 July 2023
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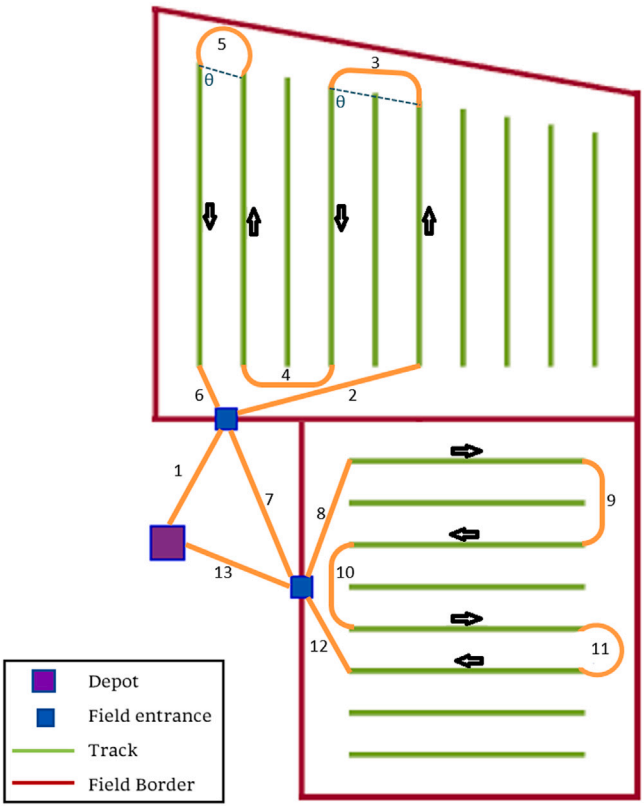


Fig. 1. Diagram of a tour and the manoeuvres of a machine in two fields.

Route, and limited allocation for the preparation of sugarcane fields were solved by Worasan et al. (2020). However, since none of the research have taken into account the in-field tracks or machines' manoeuvrings, our study focuses on these issues to provide a more efficient solution to the harvesting case.

Most research uses small to medium-size-fields with no more than 33 tracks in total (Plessen, 2019; Conesa-Muñoz et al., 2016; Bochtis et al., 2015; Jensen et al., 2015; Nilsson and Zhou, 2020). Several studies solve problems with track numbers ranging from 74 to 90 (Seyyedhasani et al., 2019; Seyyedhasani and Dvorak, 2017, 2018a). Meanwhile, rectangular and irregularly shaped fields were used by Khajepour et al. (2020), Seyyedhasani and Dvorak (2017), Zhou et al. (2015), Bakhtiari et al. (2013) and Conesa-Muñoz et al. (2016). Although irregularly shaped fields more closely resemble actual farm conditions, as stated by Zhou et al. (2015), rectangular-shaped fields remain the most commonly studied condition in ARP.

Previous studies used heuristic and metaheuristic algorithms for ARP. Among others, the heuristic algorithms used are the Clarke–Wright (Seyyedhasani et al., 2019) and greedy algorithms (Oksanen and Visala, 2009). Apart from heuristic algorithms, most research has suggested metaheuristic approaches to address the route planning problem. However, the most popular algorithms applied to the problem are the Genetic Algorithm (Evans et al., 2020; Filho et al., 2019; Nazarahari et al., 2019) and Tabu Search (Plessen, 2019; He et al., 2020; Seyyedhasani and Dvorak, 2018b). Tabu Search (TS) was used by Seyyedhasani and Dvorak (2017, 2018b) to increase field productivity and guide multiple vehicles in agricultural tasks. The findings suggest that TS could better minimise the fieldwork time than the Clarke–Wright algorithm can. Apart from the Genetic Algorithm (GA) and TS, other evolutionary and metaheuristic algorithms found in the literature include Ant Colony Optimisation (Zhou et al., 2015), Particle Swarm Optimisation (Worasan et al., 2020), Adaptive Large Neighbourhood Search (Khajepour et al., 2020), and Simulated Annealing (Conesa-Muñoz et al., 2016). These ARP studies focus on finding an optimal solution rather than shortening the runtime of their proposed method.

Several research gaps in ARP have therefore been identified. First, most of the existing studies on ARP have not taken into account the runtime of their algorithms. Therefore, the development of a fast algorithm for ARP is a promising area for further research. Additionally, our experiments revealed potential improvements for both runtime and proposed solutions for datasets published in the literature. Thus, an improved algorithm would allow for a faster runtime and superior solutions. Second, the majority of the existing studies on ARP have focused solely on single machines, disregarding capacity restrictions, manoeuvres, and multifield scenarios. Publications that have provided algorithmic details demonstrating efficient and effective solutions are typically restricted to small field sizes. Therefore, this study simultaneously addresses multiple ARP constraints and considers a large number of tracks in the problem sets.

The contribution of this study is the development of a Fast Hybrid Algorithm (FHA) to perform ARP that, considers multiple crop fields (with specific entrance points), multiple machines with the possibility of different capacities (heterogeneous machines), and machine manoeuvres. Furthermore, this study examines both medium (>25) and large (>75) numbers of tracks, whereas prior studies have typically limited the number of tracks to smaller sizes. Finally, this study conducts a series of experiments to assess the solution quality and runtime of the proposed approach compared to state-of-the-art algorithms (i.e., GA, Improved GA, ACO, and TS).

The following section describes ARP, the mathematical formula, and FHA as the proposed algorithm. Section 3 details the outcomes of the experiment, including a description of the problem sets and the parameter settings, the numerical/calculation results, and a deeper analysis. Finally, Section 4 provides some recommendations for further investigations of ARP and conclusions.

2. Materials and methods

2.1. Problem description

This study develops agricultural route planning (ARP) to cover fields with borders on a flat surface, while utilising machines with heterogeneous capacity. The problem includes field borders, which restrict access between fields, thereby preventing machines from traversing freely. In other words, the machines can only traverse through designated entrances, which serve as both entry and exit points for each field. These entrances introduce additional complexity to ARP problems, as the algorithm must consider the shortest distance from the entrances to the first and last visited tracks.

This research examines the scenario in which heterogeneous machines are utilised across multiple fields. Specifically, harvesting machines with different capacities are considered. Consequently, larger machines possess greater capacities, while smaller machines are assumed to have lower capacities. By utilising a combination of these machines, farmers and landowners can ensure the most efficient harvesting of their fields. Please refer to Section 3.2 (Table 5) for the list of the machines and their respective capacities used in this study. The capacity of each machine determines the maximum distance it can cover in a single trip. The optimisation process assigns machines to specific routes based on their capacities. Once a machine reaches its capacity, it must return to the depot to unload the collected produce before resuming operations.

Table 1
Nomenclature.

No	Notations	Description
1	V	Set of vertices inside the graph G
2	S	Subgraph of graph $G, \forall S \subseteq V$
3	i, j	Vertex index (representing the tracks); $i, j = (0), 1, \dots, n$; where 0 is the depot
4	F	Set of fields
5	a, b	Fields index; $a, b = 1, \dots, n; a, b \in F$
6	d_{ab}	The distance of two fields (between entrance of field a and entrance of field b)
7	D_{aij}	The distance from depot to the entrance of field a and to the vertex i or vertex j
8	h_a	The distance from depot to the entrance of field a
9	f_{ai}	The distance from the entrance of field a to vertex i
10	w	Effective operating width of a field's track
11	K	The set of machines (at the depot), each denoted by $k, k \in K$
12	r	Minimum turning radius of the machine
13	ψ	The driving direction of the machine
14	$g_{ij\psi}$	The length of a manoeuvre between track i and j with driving direction ψ and an angle $\theta_{ij\psi}$
15	$\theta_{ij\psi}$	The tilt angle between the track i and j and driving direction ψ, note that $0° < \theta \leq 90°$
16	$\Omega(i, j, \theta_{ij\psi})$	The bulb type of manoeuvre
17	$\Pi(i, j, \theta_{ij\psi})$	The flat type of manoeuvre
18	$\Omega\theta(i, j, \theta_{ij\psi})$	The bulb type of manoeuvre with a $\theta°$ if an angle θ is formed by the preceding and next tracks
19	$\Pi\theta(i, j, \theta_{ij\psi})$	The flat type of manoeuvre with a $\theta°$, if an angle θ is formed by the preceding and next tracks
20	u_i	The position of vertex i in a tour
21	P_i	The length (volume of crops collected) of Track i
22	Q_k	The capacity of a machine $k \in K$
23	X_{ij}^k	Decision variable, equal to 1 if machine k moves from vertex i to vertex j, otherwise $0, (i, j \in V)$

An illustration of an ARP problem with two fields and one machine is presented in Fig. 1. The orange lines represent the machine's movement that form the nonworking distance, while the black arrows indicate the machine's driving direction. Fig. 1 also demonstrates that the machine has the same starting and ending points, which are considered to be the depot. In this study, the nonworking distance comprises various distances outside the machine's tracks, including manoeuvres' length, travel between entry or exit points, and departure and entry into the depot.

Additionally, Fig. 1 depicts the trip and manoeuvres of a machine in two adjacent fields: north and south. The numbers placed next to the orange lines indicate the sequence of movements undertaken by the machine. The nonworking distances encompass manoeuvres (3, 4, 5, 9, 10, and 11), travel between the depot and entrances (1 and 13), travel from one entrance to another (7), and travel from tracks leading to the entrances (2, 6, 8, and 12).

The route begins from the depot and then passes through the north field's entrance (1–2) before visiting four tracks in the north field and performs three manoeuvres (3–5). Then, the machine leaves the northern field and enters the southern field (6–7). It starts from the first track (8), then travels to several other tracks, performs three manoeuvres (9–11), returns to the entrance (12) and returns to the depot (13). Four kinds of manoeuvres in Fig. 1 are considered in this study, which assumes the necessity of a machine to skip at least one track to achieve a flat (Π) or flatθ ($\Pi\theta$) turn. In other words, if the next track is right next to the current track, the machine needs to make a bulb(Ω) or bulbθ ($\Omega\theta$) turn. The description of each notation is provided in Table 1 (No. 16–19). In Fig. 1, the performed manoeuvres are flat (4, 9, 10), bulb (11), flatθ (3) and bulbθ (5).

Every manoeuvre's length on the different sides of the field is individually calculated with the help of the driving direction variable. Every time a machine performs a manoeuvre, it reads the coordinates of the end of the current track and the start of the next track (associated with the driving direction). The flatθ and bulbθ turns occur if the current and next tracks form an angle less than 90 degrees ($\theta < 90°$), while flat and bulb turns occur if the current and next tracks form a 90-degree angle ($\theta = 90°$). The flatθ and bulbθ turns are commonly found in irregular terrain, while flat and bulb typically occur in rectangular terrain. The conditions of these turning movements and the formulation for calculating the length of the manoeuvres are listed in Eqs. (4)–(8) in Section 2.2.

2.2. Mathematical model

Graph G shows that the vertices and edges are assumed to represent a set of tracks in the farming area or field. The parameters and variables in the mathematical model are presented in Table 1. This research uses one node representation per track, taking into account the machine's driving direction (ψ). Every time a machine passes a track, the algorithm detects its direction and position, subsequently recalculating the length of the manoeuvre.

The decision variable, X_{ij}^k is listed in the final row of Table 1. The objective function is shown in Eq. (1). The first term is the length of the manoeuvres of the machines and this is added with the second term, which is the distance from the depot to the entrance and tracks. Then, the third term is the distance between fields' entrances if a machine visits more than one field. Eqs. (2) and (3) compute the off-field distances D_{aij} and d_{ab}, respectively. D_{aij} calculates the distance when a machine goes to a field (for the first time) to visit particular track(s) or when a machine goes back to the depot from a field. Meanwhile, d_{ab} records the distance between two fields if a machine visits some tracks inside two fields.

Adjusted from equations by Jin and Tang (2010) as well as Utamima et al. (2022), Eqs. (4) through (8) in this study are utilised to compute the required range of distance the machine must travel for the purpose of making a turning movement in the headland area. In Eq. (4), $g_{ij\psi}$ is the length of a manoeuvre between tracks i and j with driving direction ψ and angle $\theta_{ij\psi}$. The condition of each type of manoeuvre is listed on the right side of Eq. (4), while the specific formulas to calculate the length of the manoeuvre($\Pi, \Omega, \Pi\theta, \Omega\theta$) are provided in Eqs. (5) through (8). Eqs. (9) and (10) ensure that each route of machines begins and ends at the depot (vertex 0). Eqs. (11) and (12) ensures that each node is visited exactly once. Meanwhile, Eq. (13) provides that a particular track visited by a machine should be the track from which the machine next departs. Eq. (14) is a subtour elimination constraint. Finally, Eq. (15) is a capacity constraint that ensures that the machine's capacity is not exceeded by the amount of harvested crop.

$$z = \min \sum_{k \in K} \sum_{i \in V} \sum_{j \in V} \left(g_{ij\psi} \cdot X_{ij}^k + X_{i=0;j}^k D_{aij} + X_{ij}^k \cdot d_{ab} \right) \tag{1}$$

s.t.:

$$D_{aij} = \begin{cases} h_a + f_{aj}, & \text{if } i = 0 \\ h_a + f_{ai}, & \text{if } j = 0 \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

$$d_{ab} = \begin{cases} d_{ab}, & \text{if prev track in field } a \text{ and next track in field } b; a \neq b \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

$$g_{ij\psi} = \begin{cases} \Pi\left(i, j, \theta_{ij\psi}\right), & \text{if } |i-j| \leq \frac{2r}{w} \wedge \theta_{ij\psi} = 90° \\ \Omega\left(i, j, \theta_{ij\psi}\right), & \text{if } |i-j| > \frac{2r}{w} \wedge \theta_{ij\psi} = 90° \\ \Pi\theta\left(i, j, \theta_{ij\psi}\right), & \text{if } |i-j| \leq \frac{2r}{w} \wedge \theta_{ij\psi} < 90° \\ \Omega\theta\left(i, j, \theta_{ij\psi}\right), & \text{if } |i-j| > \frac{2r}{w} \wedge \theta_{ij\psi} < 90° \end{cases} \tag{4}$$

$$\Pi\left(i, j, \theta_{ij\psi}\right) = |i-j|w + (\pi - 2)r \tag{5}$$

$$\Omega\left(i,j,\theta_{ij\psi}\right) = \mathrm{r}\left(3\pi - 4\sin^{-1}\left(\frac{2\mathrm{r}+|i-j|w}{4r}\right)\right) \tag{6}$$

$$\Pi\theta\left(i,j,\theta_{ij\psi}\right) = |i-j|w\left(1+\cot\theta_{ij\psi}\right) + r(\pi-2) \tag{7}$$

$$\Omega\theta\left(i,j,\theta_{ij\psi}\right) = \pi r + \frac{4r^2 - |i-j|w\left(4r + w\cot^2\theta_{ij\psi} + w\right)}{4r - 2w|i-j|} \times \sin^{-1}\frac{|i-j|w\left(4r\cot\theta_{ij\psi} - 2w\cot\theta_{ij\psi}\right)}{4r^2 - |i-j|w\left(4r + w\cot^2\theta_{ij\psi} + w\right)} \tag{8}$$

$$\sum_{j\in\mathrm{V}} X_{0j}^k = 1 \quad ,\forall k \in \mathrm{K} \tag{9}$$

$$\sum_{i\in\mathrm{V}} X_{i0}^k = 1 \quad ,\forall k \in \mathrm{K} \tag{10}$$

$$\sum_{k\in K}\sum_{i\in V} X_{ij}^k = 1 \quad i,j \neq 0, \forall j \in V \tag{11}$$

$$\sum_{k\in K}\sum_{j\in V} X_{ij}^k = 1 \quad i,j \neq 0, \forall i \in V \tag{12}$$

$$\sum_{k\in K}\sum_{i\in V} X_{ij}^k = \sum_{k\in K}\sum_{j\in V} X_{ji}^k \quad i,j \neq 0, \forall j \in V \tag{13}$$

$$u_i - u_j + Q_k P_i X_{ij}^k \leq Q_k - 1 \quad i,j \neq 0, \forall i,j \in V, \forall k \in K \tag{14}$$

$$\sum_{i\in V}\sum_{j\in V} P_i X_{ij}^k < Q_k \quad \forall k \in K \tag{15}$$

2.3. Experimental methods

2.3.1. Proposed method: Fast hybrid algorithm

The current study develops a new hybrid approach called Fast Hybrid Algorithm (FHA). FHA combines a population-based concept, a fast neighbourhood search, a local search approach, and an elitism technique. The proposed method, FHA, does not make specific assumptions but rather attempts to find the best solution. FHA considers the route and turn optimisation simultaneously, and therefore, instead of looking at a tour, it identifies the best turn (or vice versa). FHA determines a pool of candidate solutions in each iteration and calculates the route's distances. The best route is identified at the end of the algorithm's process. The algorithms used in the authors' earlier papers, Utamima et al. (2022), is improved with FHA, resulting in faster running time while maintaining effective solutions.

Generally, FHA has two primary stages: exploration and exploitation. The former stage searches for a better solution from the individuals (candidate solution) within the population, while the latter stage aims improve the current best solution. FHA employs a fast neighbourhood search that uses six combinatorics operators in the exploration of neighbourhood areas:

1. Swap two : swap two tracks in a candidate solution with the other two tracks inside it.
2. Flip : flip part of the tracks in a candidate solution.
3. Interchange : interchange two tracks randomly in a candidate solution.
4. Slide : move a track into a position and slide the other tracks forward or backward.
5. Guided swap : place a track in the corresponding position for one with the same global best position.
6. Scramble : randomly reorder the tracks in a candidate solution.

Algorithm 1 FHA's pseudocode

```
1: procedure FHA()
2:     Initialisation ReadData()
3:     for g = 1 to maxGeneration do
4:         ObF ← CalculateObjectiveFunction(Population)
5:         for i=1 to popSize do
6:             S ← RouletteWheel()
7:             if rand < 0.9 then
8:                 [T] ← RandomSelect two points()
9:                 a ← Rand(5)
10:                switch a do
11:                    case 1
12:                        NewCandidates ← SwapTwo(S, T)
13:                    case 2
14:                        NewCandidates ← Flip(S, T)
15:                    case 3
16:                        NewCandidates ← Interchange(S, T)
17:                    case 4
18:                        NewCandidates ← Slide(S, T)
19:                    case 5
20:                        NewCandidates ← GuidedSwap(S, T)
21:                end switch
22:            else
23:                NewCandidates ← Scramble(S)
24:            end if
25:        end for
26:        ObF ← Calculate ObjectiveFunction(NewCandidates)
27:        crnBest ← Min(ObF)
28:        if crntBest < globalBest then
29:            gloBest ← crnBest
30:        else
31:            lsSol ← gloBest
32:            while loop < maxLoop do
33:                lsSol ← Swap
34:            end while
35:            if lsSol > globalBest then
36:                while loop < maxLoop do
37:                    lsSol ← Slide
38:                end while
39:            end if
40:        end if
41:        if lsSol < gloBest then
42:            gloBest ← lsSol
43:        end if
44:        DoElitism()
45:        Population ← NewCandidates
46:    end for
47: end procedure
```

Algorithm 1 lists the FHA pseudocode. In Algorithm 1, first, the FHA parameters are initialised. The algorithm reads the field data (the coordinates of tracks, entrances, and depot). Next, the candidate solutions are generated, and their objective function values are calculated.

The main iteration starts with the exploration stage using the combinatorics operator (Algorithm 1, Lines 5–25). The candidate solutions (S) for the next population are selected using a roulette wheel selection method that applies a proportional chance to be selected to the fitness of the candidate solution.

In the exploration stage, the application of the combinatorics operator is differentiated by the rate. Preliminary experiments have shown that while scramble operators should be executed with a 10% probability, all other operators are optimally set at 18% (90% in total for five operators). The algorithm randomly selects two points from S and applies the combinatorics operators to the candidate solutions (Algorithm 1, Lines 8–21). The result of the exploration stage is the new population.

In the next step, the fitness value of each individual in the new population is computed (Algorithm 1, Line 26) so that the current best

solution (*crnBest*) is then found (Algorithm 1, Line 27). The *crnBest* is analysed if it satisfies the criteria of global minimum to replace the global best solution (Algorithm 1, Lines 28–29). If *crnBest* is not superior to the global best solution (*gloBest*), then the exploitation stage will be executed. The exploitation stage uses an improved local search that employs the swap and slide operators until no further improvement can be made (Algorithm 1, Lines 31–39).

On the other hand, if the solution of the local search (*lsSol*) is better than *gloBest*, then *lsSol* will become *gloBest* (Algorithm 1, Lines 41–43). Next, the elitism procedure is applied to maintain 10% of the best candidate solutions through each generation to save potential solutions (Natesha and Guddeti, 2021). The recorded candidate solutions are placed into the new population, which is used as the initial population for the next generation. In this research, FHA is developed in MATLAB.

2.3.2. Published route planning methods (Comparer)

In agricultural route planning, some metaheuristic methods have been commonly used. Among the metaheuristic methods, Genetic Algorithm (GA), Ant Colony Optimisation (ACO), and Tabu Search (TS) are employed in this research. Additionally, we develop an improved version of GA to include as a comparison algorithm.

TS is an algorithm that traverses the solution space by going iteratively from the present solution to the best (neighbouring) solution. The TS procedure starts with a feasible solution and iteratively advances towards another solution until a near-optimal solution is reached. TS enhances the search by not re-evaluating previously visited points in the search space. Such a point is regarded as "taboo" and must be reviewed again. This method employs flexible memory structures with constraints and aspiration criteria for exploiting the search spaces (Glover and Laguna, 1997). TS is used in ARP applications to optimise the nonworking distance for vehicles by identifying a field's boundaries and centre (He et al., 2020). Plessen (2019) also employed TS to reduce the nonworking distance in full and partial field coverage on irregularly shaped fields. The authors emphasised the use of the tabu list in TS to improve algorithm exploration and reduce computation runtimes.

ACO is a metaheuristic method inspired by the pheromone-trail behaviour of real ant colonies. As a guide for their nest-mates to the food source, the ants designate the path back to the nest with a pheromone, a chemical indicator, as they return through the forest after feeding. Individual ants may use distinct routes to reach the same food source. However, the shortest path will accumulate the maximum concentration of pheromone, thereby increasing the likelihood of selection. A group of agents, or artificial ants, search for optimal solutions to a given optimisation problem (Barbosa, 2013). In ARP applications, Jing et al. (2022) use improved ACO to optimise path for autonomous land levelling operations. Zhou et al. (2015) applied ACO to a problem involving a single field with obstacles. The authors demonstrated that ACO could generate superior solutions to farmers' manual labour.

Meanwhile, GA is applicable to a wide variety of optimisation problems, including ARP. GA is based on evolution and is applied to the population of chromosomes (potential candidate solutions). In general, the algorithm begins with the generation of a new population and then selects chromosomes with higher fitness values. The better the fitness of a chromosome, the greater its possibility of being selected. With a specified probability, crossover and mutation are applied to the chosen chromosomes. The crossover process combines genes from the parents to create new chromosomes. The mutation procedure modifies the value of genes within a few chromosomes. In the context of ARP, GA was used to optimise the path of the row crop grain harvester (Evans et al., 2020). Meanwhile, Nazarahari et al. (2019) solved the route planning of multiple mobile robots using an extended GA.

In this study, the basic GA's crossover and mutation use one operator each. The performance of GA can be improved by adding more operators in its structure (Nazarahari et al., 2019). Thus, we also develop an improved version of GA that uses two kinds of operators (one-point and two-points) in the crossover and three kinds of operators (flip, interchange, and slide) in the mutation part. This improved version of GA is called Improved GA (IGA).

3. Experimental results and analysis

3.1. Comparison with published dataset

There are published datasets of agricultural route planning gathered in Utamima et al. (2019). The datasets were sourced from research publications and their respective authors. Table 2 lists the summary of the data. The dataset contains rectangular and irregular fields. The first column of Table 2 lists the problem code. The 'Track Width & Turning Radius' column contains the width of the tracks and the turning radius of the machine; the 'Type of Manoeuvre' column contains the types of turns that are utilised in this particular dataset. The 'Machine' column details the number of used machines (note that all cases use the homogeneous type of machine(s)), while the last column lists the references for the datasets.

This research tests the proposed algorithm (FHA) against published data (Table 2). Table 3 shows the solution obtained by FHA on the published datasets. The problem code is shown in the first column of Table 3, and the best-known solution is shown in the second column, which is taken from the most recent literature (Seyyedhasani and Dvorak, 2018a; Conesa-Muñoz et al., 2016). Column 4 lists the results achieved by FHA. The values in bold from Columns 2 and 4 refer to the smallest value (minimum distance) found in that row.

The experimental results show that FHA produces equally optimal, best-known solutions for six instances (8rt, 12rt_a, 12rt_b, 20rt, 62Hs, 74nc2), and achieves better solutions for the other three cases (37nc, 90nc, 74nc). Accordingly, the capability of FHA to exceed the anticipated results for the ARP datasets is shown due to FHA's efficient structure while following the exploration and exploitation strategy.

3.2. Test with multiple fields and multiple machines problems

The present study performs experiments that apply the proposed method (FHA/Fast Hybrid Algorithm) to the ARP problem, and comparative experiments of the existing algorithms with FHA. Additionally, this research recoded the existing algorithms (i.e., GA/Genetic Algorithm, ACO/Ant Colony Optimisation, and TS/Tabu Search) with the same objective function as FHA. Additionally, the Improved GA (IGA) is also developed as comparison algorithm.

In Table 4, the characteristics of the ten datasets (problems) utilised in this research are listed. Ten problems are presented and the total number of tracks, track lengths, and shapes vary. The number of fields can be up to five, while the total number of tracks ranges from 28 to 112. Fig. 2 illustrates the samples of problem cases. Several problems, such as P4c, include a special case regarding the field's entrance where a machine needs to pass through the south field before reaching the entrance of the north field. Hence, for efficiency, the optimisation needs to consider performing fieldwork on the tracks of the south field before or after going to the north field.

Seven machines with heterogeneous capacities are available and shown in Table 5 (ranging from 1000 to 3000 m). As previously mentioned, this research assumes the same amount of harvest per travelled distance. Therefore, the maximum distance a machine can travel in a single ride serves as an illustration of its capacity. Each machine also needs to start from and return to the depot.

Table 6 displays the experimental findings (measures) regarding the nonworking distance of several fields with multimachine ARP problems. Columns one and two list the problem code and the number of tracks, respectively, and the remaining columns contain the solutions achieved by GA, IGA, ACO, TS and FHA in order. The values highlighted in bold correlate to the row's minimum value. The distance savings achieved by FHA when compared to GA (Column 3), IGA (Column 4), ACO(Column 5), and TS (Column 6) are listed in the very last row of Table 6.

As shown in Table 6, compared to the other methods, FHA consistently obtains the shortest nonworking distance in all problems.

Table 2
The details of published ARP datasets.

Problem code	Track Width & Turning Radius	Shape	Type of Manoeuvre	Machine (homogeneous)	References
8rt 12rt_a	2.89 & 3.5	Rectangular	Flat, bulb	1	Bochtis and Vougioukas (2008)
12rt_b 20rt	2.5 & 3.5	Rectangular	Flat, bulb	1	Conesa-Muñoz et al. (2016)
37nc	9 & 6	Irregular	Flat, bulb, flatΘ	1	Conesa-Muñoz et al. (2016); Hameed et al. 2011
62Hs	2.5 & 2.3	Irregular	Flat, bulb	1	Oksanen and Visala (2009)
90rt	10 & 7	Rectangular	Flat, bulb	3	Seyyedhasani and Dvorak (2017)
74nc2	19 & 9.77	Irregular	Flat, bulb, flatΘ, bulbΘ	3	Utamima et al. 2019b
74nc	19 & 9.77	Irregular	Flat	3	Seyyedhasani and Dvorak (2018a)

Table 3
Test with published ARP datasets.

Problem	Best-known solution	References	Proposed algorithm: FHA
8rt	**94.439**		**94.439**
12rt_a	**146.027**	Bochtis and Vougioukas (2008) and	**146.027**
12rt_b	**145.602**	Conesa-Muñoz et al. (2016)	**145.602**
20rt	**235.491**		**235.491**
37nc	961.470	Conesa-Muñoz et al. (2016)	**958.930**
62Hs	**479.914**	Utamima et al. (2019b)	**479.914**
90rt	2658.474	Utamima et al. (2019b)	**2642.72**
74nc2	**5197.349**	Utamima et al. (2019b)	**5197.349**
74nc	4416.300	Seyyedhasani and Dvorak (2018a)	**3832.206**

Table 4
ARP problems' description.

No.	Problem code	Total tracks	Range of track lengths	Track width	Shape
1	P2a	28	152–181	5	Irregular (polygon with 4–5 sides)
2	P2b	30	153–205	5	Irregular (polygon with 4 sides)
3	P2c	36	30–152	5	Irregular (polygon with 4 sides)
4	P3a	56	40–150	5	Irregular (polygon with 4 sides) and Rectangular
5	P3b	62	30–195	5	Irregular (polygon with 4 sides)
6	P4a	78	10–150	4	Irregular (polygon with 4 sides) and Rectangular
7	P4b	86	47–115	4	Irregular (polygon with 4–6 sides) and Rectangular
8	P4c	100	13–149	4	Irregular (polygon with 4 sides) and Rectangular
9	P5a	104	52–109	4	Irregular (polygon with 4–6 sides)
10	P5b	112	15–200	4	Irregular (polygon with 4 sides)

Table 5
Machines provided.

Machine (heterogeneous)	Maximum working distance
A	3000
B, C, D	2500
E, F	2000
G	1000

Accordingly, FHA outperforms GA, IGA, ACO and TS. Furthermore, the solutions of GA and IGA are superior to TS in P2c, P3b, P4c and P5b, while the solutions of TS are better than those produced by both GA and IGA in the other six cases (P2a, P2b, P3a, P4a, P4b and P5a). Meanwhile, ACO's solutions are the worst compared to the rest of the algorithms.

When comparing FHA and GA, it is observed that during each iteration of the exploration phase, FHA selects one of five types of combinatorial operators, whereas GA employs one-point crossover operators. In one-point crossover, a 'crossover point' is randomly selected on both chromosomes of the parents. Subsequently, the bits located to the right of this point are interchanged between the two parent chromosomes. According to Prügel-Bennett (2001), the one-point crossover slowly mixes the population which may lead to slow exploration skills. In contrast, FHA has five various operators that strengthen its exploration ability. Meanwhile, although the improved GA uses more operators than GA, it does not embed a local search similar to that in FHA; therefore, the IGA still lacks the exploitation ability that makes the IGA's solution unable to catch with FHA.

The artificial ants in ACO utilise a randomised construction heuristic that uses artificial pheromone trails to make probabilistic decisions to solve problems that depend on the input data. Although ACO has the capability to search globally and converge quickly towards optimal solutions, it has certain limitations such as a low ability to scatter the population and a lack of a systematic method for starting up (Tiwari and Jain, 2015). This drawback of ACO is often overcome by hybridising it with other algorithms, such as a local search algorithm. Thus, comparing ACO and FHA, the performance of FHA is better than ACO because FHA includes a local search in its structure. Meanwhile, the TS solution is not better than FHA because it only uses one kind of operator (swap) in its structure.

Table 7 presents the details of the solution generated by FHA for several problems. While the first column displays the problem code, the second column indicates the utilised machines, and the last column shows the field's label and the tracks' order corresponding to Fig. 2. For instance, P2c requires two machines (where the maximum distance to harvest is 3000 m and 2500 m, respectively). Machine A travels to the first field and visits 16 tracks in the following order: 2, 1, 7, 8, 9, 10, 11, 13, 15, 16, 18, 17, 14, 12, 9, and 6, makes its way to the second field and visits four tracks in the following order: 32, 34, 30 and 31, and then goes back to the first field and visits track 3. Machine B visits track 4 of Field 1 and moves to Field 2 to visit fourteen tracks and then goes back to Field 1 and visits track 5.

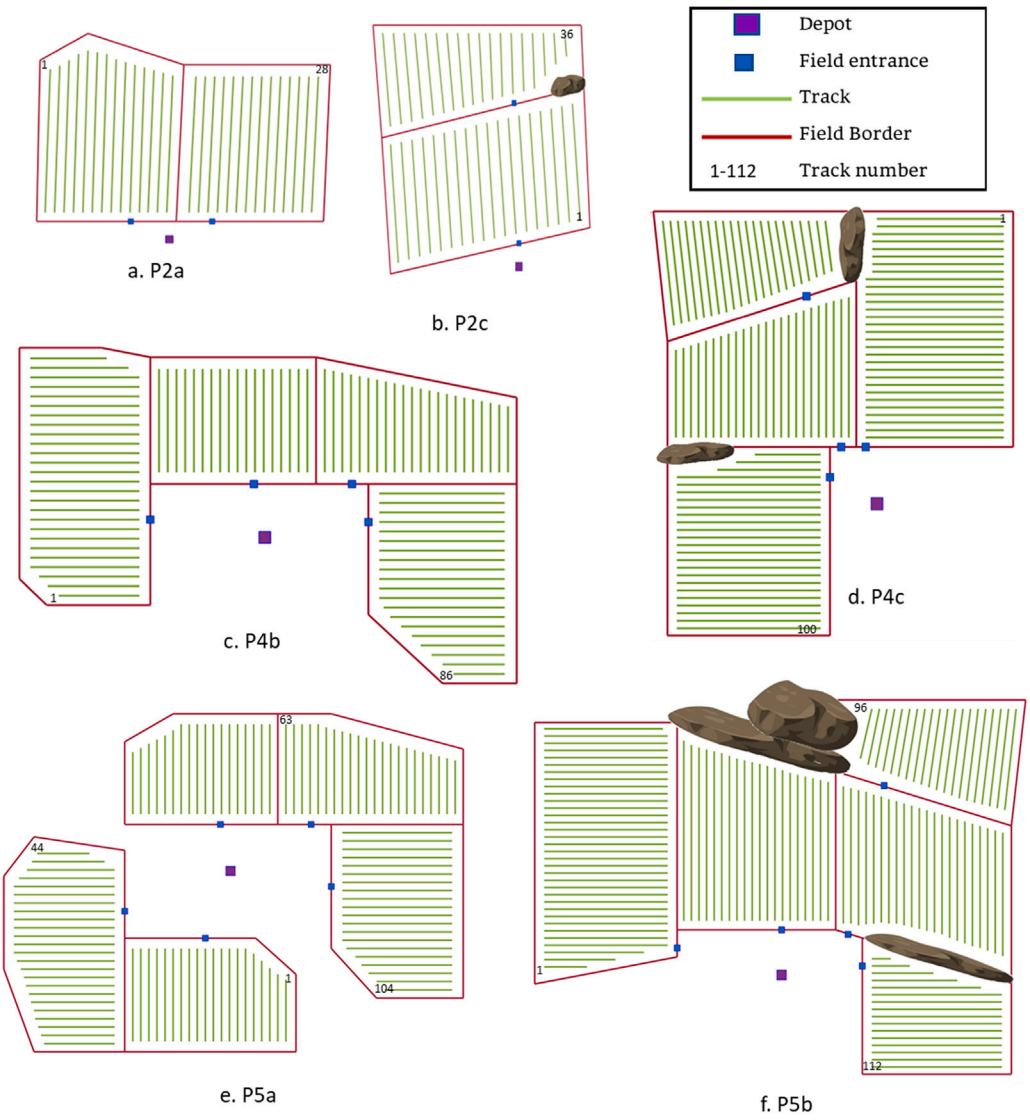


Fig. 2. Illustration of the ARP problems.

Table 6
The results of the proposed method (FHA) and other algorithms in solving ARP.

Problem code	Number of tracks	GA	IGA	ACO	TS	Fast Hybrid Algorithm (FHA)
P2a	28	798.954	782.271	799.842	774.675	**722.415**
P2b	30	703.969	695.931	709.144	689.376	**635.551**
P2c	36	897.469	892.751	913.769	906.469	**835.015**
P3a	56	1520.276	1503.849	1540.052	1127.154	**1068.886**
P3b	62	1357.039	1329.742	1490.972	1461.004	**1236.504**
P4a	78	1749.986	1721.466	1840.021	1616.605	**1553.498**
P4b	86	1814.678	1783.642	1887.351	1759.873	**1624.798**
P4c	100	2947.510	2926.431	3190.253	3083.430	**2157.403**
P5a	104	2678.439	2649.532	2740.092	2604.086	**2441.982**
P5b	112	4287.438	4237.074	4640.578	4532.694	**3896.531**
FHA′ distance savings		16.149%	14.641%	21.419%	12.613%	

Figs. 3 and 4 depict the manoeuvres of the machines in P3b and P5a, respectively. Each field is labelled by a number in the middle, while the track labels are shown by a number in each field's corner. Each of the machine tours is colour coded by lines. Each tour starts from and returns to the depot. The track widths in P3b and P5a iare 5 and 4 metres, respectively. The machine's turning radius is 3 metres in both problems. Therefore, according to Eq. (4) in Section 2.2, a flat turn can be made by the machine if at least one track is skipped before moving on to the following track.

In Fig. 3, three machines need to visit all fields in P3b. The tours of Machines A, B, and C are shown with orange, blue, and purple lines respectively. All machines successfully manage the shortest path when travelling from and to the entrance. In Fig. 3, there is a special case in which the entry of Field 3 is north of Field 2; hence, the machine can

Table 7
The details of solution in several problem by FHA.

Problem	Machine	Field number [sequence of tracks]
P2c	A	Field 1 [2, 1, 7, 8, 9, 10, 11, 13, 15, 16, 18, 17, 14, 12, 9, 6]; Field 2 [32, 34, 30, 31]; Field 1 [3];
	B	Field 1 [4]; Field 2 [29, 28, 27, 25, 24, 22, 21, 19, 20, 23, 26, 36, 35, 33]; Field 1 [5]
P3b	A	Field 1 [15, 16, 19, 20, 18, 17, 14, 13, 10, 9, 6, 5, 2, 1, 3, 4, 7, 8, 11, 12]
	B	Field 2 [24]; Field 3 [57, 54, 53, 50, 49, 46, 45, 43, 44, 47, 48, 51, 52, 55, 56, 58, 59, 61, 62, 60] Field 2 [23, 21, 22]
	E	Field 2 [25, 28, 31, 32, 35, 36, 39, 40, 42, 41, 38, 37, 34, 33, 30, 29, 27, 26]
P4c	A	Field 1 [30, 28, 29, 27, 25, 23, 21, 20, 18, 16, 14, 12, 10, 8, 6, 4, 1, 3, 2, 5, 7, 9, 11, 13, 15, 17, 19, 22, 24, 26]
	B	Field 2 [31, 32, 39, 41, 43, 45, 47, 49, 53, 54, 51, 52, 50, 48, 46, 44, 42, 40, 38, 34]
	C	Field 2 [33] Field 3 [73, 75, 76, 74, 72, 67, 65, 62, 59, 56, 55, 57, 58, 60, 61, 63, 64, 66, 68, 70, 69, 71]; Field 2 [37]
	D	Field 2 [36, 35] Field 4[81, 86, 88, 91, 93, 95, 97, 99, 100, 98, 96, 94, 92, 90]
	E	Field 4 [79, 82, 84, 89, 87, 85, 83, 77, 78, 80]
P5a	A	Field 1 [12, 14, 16, 18, 20, 19, 17, 15, 13, 11, 9, 7, 5, 2, 1, 3, 4, 6, 8, 10] Field 2 [36, 34, 32, 31, 29, 30, 27, 28, 26, 25, 23, 24, 22, 21]
	B	Field 2 [37, 38, 40, 43, 44, 42, 41, 39, 35, 33]; Field 3 [51, 50, 48, 47, 45, 46, 49, 52, 54, 57, 59, 61, 62, 60, 58, 56]
	C	Field 4 [65, 63, 64, 66, 68, 70, 73, 75, 77, 78, 80, 82, 84, 83, 81, 79, 76, 74, 72, 71, 69, 67] Field 5 [93, 97, 99, 101, 102, 104, 103, 100, 92, 94]
	E	Field 5 [96, 98, 95, 90, 87, 85, 88, 86, 89, 91]
P5b	A	Field 1 [30, 26, 24, 18, 15, 13, 7, 5, 3, 1, 2, 4, 6, 9, 12, 10, 8, 11, 14, 16, 21, 25, 28, 32, 34, 33]
	B	Field 1 [31, 29, 27, 22, 20, 17, 19, 23]; Field 2 [46, 45, 50, 49, 51, 52, 54, 53]
	C	Field 2 [48, 44, 42, 40, 38, 36, 35, 37, 39, 41, 43, 47];
	D	Field 3 [59]; Field 4 [93, 95, 96, 94, 92, 90, 88, 86, 84, 82, 80, 78, 77, 79, 81, 83, 85, 87, 89, 91] Field 3 [58]
	E	Field 3 [55, 56, 69, 71, 73, 75, 76, 74, 72, 70, 68, 61]
	F	Field 3 [57, 63, 65, 67, 66, 64, 62, 60]]; Field 5 [97, 99, 104, 107, 109, 111, 110, 112, 98]
	G	Field 5 [98, 101, 103, 105, 108, 106, 102, 100]

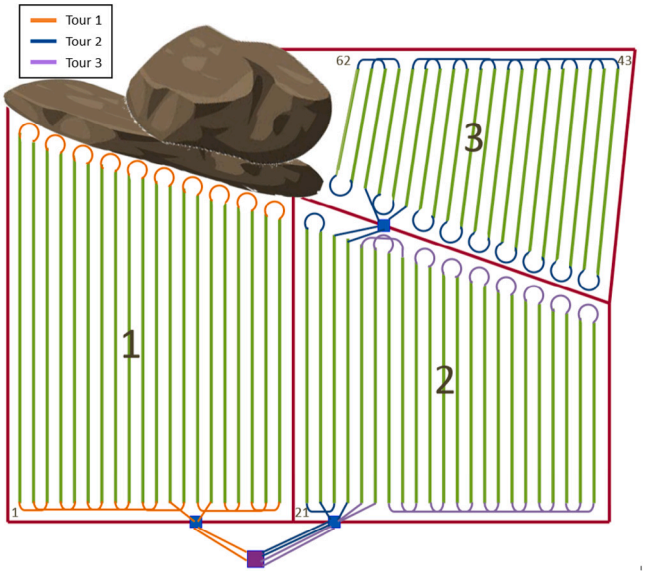


Fig. 3. Illustration of the machines' routes in P3b.

enter Field 3 after passing Field 2. FHA successfully solves this case by producing the route for Machine B (blue colour). Machine B visits one track in Field 2 and then moves to the north to harvest all tracks in Field 3, goes to Field 2, visits three tracks in Field 2, and goes back to the depot. Meanwhile, Machine C goes to the remaining tracks in Field 2 and Machine A visits all tracks in Field 1.

Fig. 4 illustrates the tours of four machines in a problem comprising five fields and 104 tracks. Machine A visits all tracks in Field 1, moves to Field 2, and visits 14 tracks in Field 2. Machine B visits the remaining tracks in Field 2, moves to Field 3, and harvests all tracks in Field 3. Meanwhile, Machine C visits all tracks in Field 4, moves to Field 5, and visits 8 tracks in Field 5. Next, Machine D harvests the rest of the tracks in Field 5. The predominant kind of manoeuvre, as seen in Fig. 4, is a flat turn.

Table 8 lists the runtime of the proposed algorithm against literature-referenced algorithms. The problem code is listed in the first column, while the remaining columns list the runtimes of GA, IGA, ACO, TS and FHA. Meanwhile, the last row of Table 8 provides the reduction in time from FHA (to produce solution) compared to GA, IGA, ACO, and TS. The value of the runtime savings of FHA is calculated with Eq. (16), where t_a^i is the other algorithm's runtime in Problem i, t_{fha}^i refers to FHA's runtime in Problem i, and n is the total number of problem cases. FHA's runtime savings are more than half of that on GA, IGA and TS, and almost half of that on ACO (last row of Table 8).

$$\frac{\sum_{i=1}^{n} \frac{t_a^i - t_{fha}^i}{t_a^i}}{n} \times 100\% \tag{16}$$

FHA has a faster runtime than all of the other algorithms (GA, IGA, ACO and TS). As the size of the problem increases, the gap between FHA's runtime and those of the others becomes larger. Meanwhile, the runtime of TS is faster than that of GA in seven problems, while in three problems, TS's runtime is slower than that of GA. Although ACO's solutions are the weakest, its runtime is faster than that of GA, IGA and TS. Nevertheless, FHA remains the fastest algorithm since FHA's runtime, on average, is more than two times faster than that of GA, IGA, ACO and TS.

Comparing the runtime of FHA with that of both GA and IGA, the experiments in this study also reveal that the crossover operators run slower than FHA's operators which makes the running times of both GA and IGA slower than that of FHA. In general, crossover may take more time than FHA operators because it involves selecting two chromosomes (candidate solutions) and combining their genetic information (Singh and Gupta, 2022), while FHA operators only involve changing genes in a single chromosome.

The ability of FHA to achieve excellent results on the multifield and multimachine dataset is because FHA incorporates various combinatorics operators in its exploration strategy, while also exploiting every best found solution in each iteration. The fast computation time of FHA is because of the efficient structure of the algorithm.

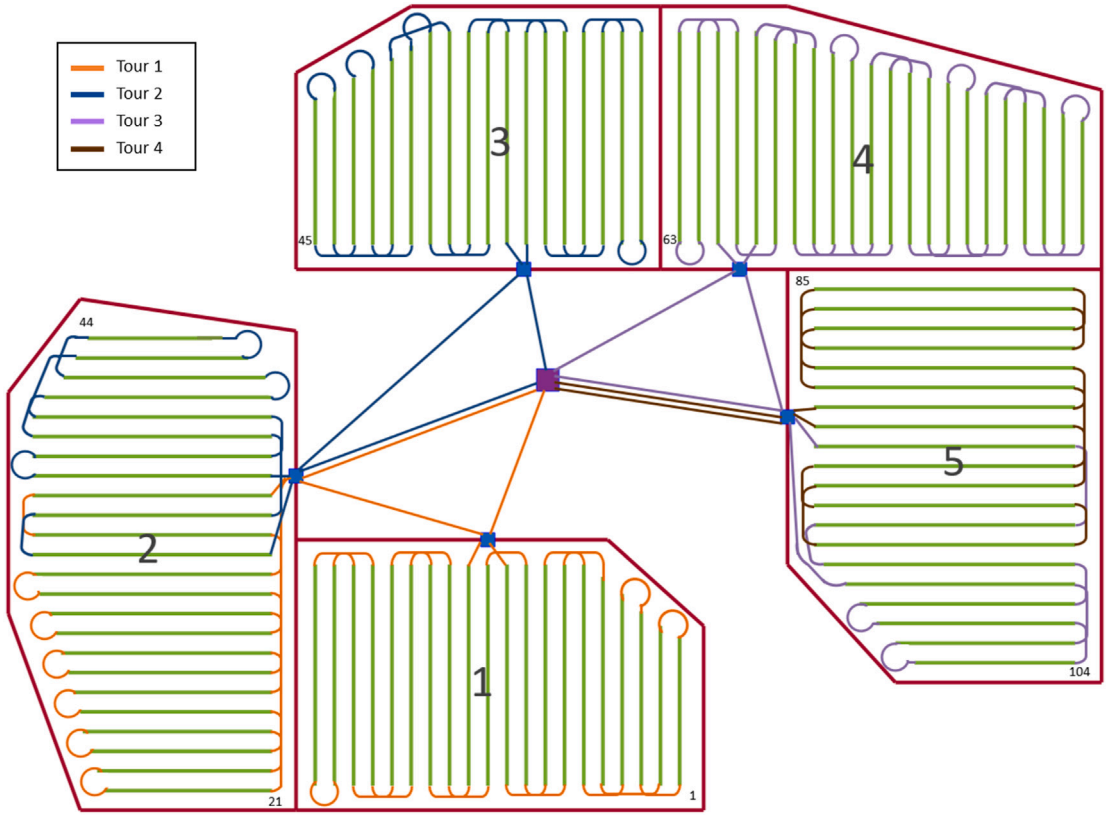


Fig. 4. Illustration of the tours in P5a.

Table 8
The average runtime (in seconds) of the proposed method (FHA) and other algorithms in solving ARP.

Problem code	Genetic Algorithm (GA)	Improved GA (IGA)	Ant Colony Optimisation (ACO)	Tabu Search (TS)	Fast Hybrid Algorithm (FHA)
P2a	1.17	1.17	1.12	1.13	**0.56**
P2b	1.38	1.39	1.29	1.35	**0.75**
P2c	6.84	6.98	6.32	9.171	**3.08**
P3a	4.14	4.27	3.87	2.94	**1.81**
P3b	12.26	12.91	13.15	16.98	**4.95**
P4a	7.39	7.54	6.539	6.642	**3.63**
P4b	6.98	7.06	6.89	7.367	**4.52**
P4c	25.61	26.75	23.13	23.81	**7.26**
P5a	9.89	10.14	8.03	8.16	**5.78**
P5b	26.44	27.43	25.28	25.63	**9.18**
FHA′ runtime savings	53.32%	54.23%	49.76%	51.76%	

4. Conclusion

In this work, a Fast Hybrid Algorithm (FHA) is developed to address various attributes to the ARP and showcase enhancements in both quality and, more importantly, the time required to solve problems. By producing better routing designs, this research is expected to improve the route planning of (semi-) autonomous agricultural machines.

This study encompasses agricultural route planning which involves multiple fields utilising machines with varying capacities on a flat surface. From the solutions obtained through the Fast Hybrid Algorithm (FHA), it is observed that the machines not only select efficient paths for manoeuvring but also achieve the shortest route when travelling to and from the entrance. Moreover, when solving ten ARP problem sets, FHA achieves the best objective function, resulting in savings of up to 21% compared to other algorithms previously reported in the literature. Additionally, FHA exhibits the fastest execution time for all the given problems, highlighting its efficiency. On average, FHA's running time is more than twice as fast as that of the GA, IGA, ACO, and TS algorithms. Therefore, this study recommends the implementation of FHA in ARP, as it outperforms the GA, IGA, ACO, and TS algorithms.

The contribution of this research is considered a foundation for real-world problem applications and a starting point to extend to other attributes or algorithmic improvements. Future research can concentrate on applying different constraints to ARP with variations in field layouts and conditions. Currently, FHA can solve problems that contain eight to 112 tracks on fields with flat surfaces. The range and the problem type are the current limitations of FHA since further research on applying larger tracks and under various field conditions is needed to test the performance of FHA. Another research direction for FHA is applying the algorithm to different optimisation of combinatorial problems.

CRediT authorship contribution statement

Amalia Utamima: Conceptualization, Methodology, Software, Writing – original draft, Visualization. **Torsten Reiners:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors gratefully acknowledge financial support from the Institut Teknologi Sepuluh Nopember for this work, under project scheme of the Publication Writing and IPR Incentive Program (PPHKI) 2023.

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